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Intracellular Spermine Is a Key Player in GSG1L's Regulation of Calcium-Permeable AMPAR Channel Conductance and Recovery from Desensitization

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Calcium-permeable AMPA-type glutamate receptors (CP-AMPARs) contribute to excitatory synaptic transmission and play pivotal roles in normal and detrimental forms of plasticity. Most AMPARs are associated with auxiliary subunits. Transmembrane AMPAR regulatory proteins (TARPs), the prototypical auxiliary subunits, enhance current through CP-AMPAR channels by increasing single-channel conductance and relieving the characteristic voltage-dependent block by endogenous intracellular polyamines, such as spermine. In contrast, the atypical auxiliary subunit GSG1L negatively regulates CP-AMPARs, suppressing current flow by promoting polyamine block and reducing channel conductance. Here, we investigated the role of polyamines in these opposing effects. We show that, in the absence of auxiliary subunits, intracellular spermine decreases CP-AMPAR single-channel conductance. This effect is prevented by the presence of TARPs but enhanced by the presence of GSG1L. Unexpectedly, intracellular spermine is necessary for both GSG1L's attenuation of CP-AMPAR channel conductance and its characteristic slowing of recovery from desensitization. These various effects are determined by specific residues within the channel's selectivity filter and within GSG1L's C-tail. Together, our findings reveal that intracellular polyamines play an essential role in GSG1L's unique ability to negatively regulate many of the key properties of CP-AMPARs.

Key words: AMPA receptor auxiliary subunits; desensitization; GSG1L; polyamines; single-channel; TARPs

Significance Statement

Calcium-permeable AMPA receptors (CP-AMPARs) are key players in synaptic transmission and plasticity. Our study reveals a novel mechanism by which intracellular polyamines, in conjunction with auxiliary subunits, negatively modulate the function of recombinant CP-AMPARs. We show that spermine decreases CP-AMPAR single-channel conductance, an effect that is regulated differentially by transmembrane AMPAR regulatory protein and GSG1L auxiliary subunits. Importantly, our study uncovers an unexpected requirement for intracellular spermine in mediating GSG1L's effects on key CP-AMPAR properties, including channel conductance and recovery from desensitization, and provides new insight into the complex interplay between CP-AMPARs, their auxiliary subunits, and endogenous intracellular polyamines.

Introduction

Intracellular polyamines, such as spermidine and spermine, are polycationic molecules that act as important regulators of neuronal excitability by producing voltage-dependent block of a wide range of cation-selective ion-channel families. These include inward rectifier K^+ channels (Lopatin et al., 1994; Oliver et al., 2000), cyclic nucleotide-gated channels (Lu and Ding, 1999; Guo and Lu, 2000), BK channels (Zhang et al., 2006), TRPC channels (Kim et al., 2016), TRPV channels (Maksaev et al., 2023), voltage-gated Na^+ channels (Fleidervish et al., 2008), neuronal nicotinic acetylcholine receptors (Haghighi and Cooper, 1998, 2000), and calcium-permeable AMPA- (CP-AMPAR) and kainate-type glutamate receptors (Bowie and Mayer, 1995; Kamboj et al., 1995; Twomey et al., 2018).

AMPARs are critical for normal brain function as mediators of fast excitatory transmission (Henley and Wilkinson, 2016; Hansen et al., 2021). These ligand-gated channels can function either as homo- or heterotetrameric assemblies of GluA1-4 pore–forming subunits. For the GluA2 subunit, posttranscriptional RNA editing results in a switch from a neutral glutamine to a positively charged arginine at the "Q/R site" in the AMPAR channel's ion selectivity

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filter, rendering GluA2-containing receptors calcium-impermeable (CI-AMPARs) and insensitive to cytoplasmic polyamines (Sommer et al., 1991; Verdoorn et al., 1991; Burnashev et al., 1992; Kuner et al., 2001). In contrast, AMPARs that lack GluA2 are calcium permeable and characterized by currents that are inwardly rectifying (Bowie and Mayer, 1995; Kamboj et al., 1995; Koh et al., 1995; Bahring et al., 1997). This latter feature results from the presence of intracellular polyamines that act as weakly permeable channel blockers (Bowie and Mayer, 1995; Brown et al., 2018). Such CP-AMPARs play a pivotal role in both normal and maladaptive (disease-related) forms of synaptic plasticity (Cull-Candy and Farrant, 2021).

Native AMPARs coassemble with various transmembrane auxiliary proteins that not only influence their biogenesis and synaptic targeting but also modify their gating and pharmacology (Jackson and Nicoll, 2011; Greger et al., 2017; Schwenk et al., 2019; Kamalova et al., 2020; Certain et al., 2023). Of the core auxiliary proteins, the transmembrane AMPAR regulatory proteins (TARPs) and cornichons have both been found to partially relieve the block of CP-AMPARs by intracellular polyamines (Cho et al., 2007; Soto et al., 2007; Coombs et al., 2012; Brown et al., 2018), by increasing the rate of blocker permeation (Brown et al., 2018). Additionally, they induce a characteristic increase in single-channel conductance and slowing of channel gating that is independent of intracellular polyamines (Tomita et al., 2005; Coombs et al., 2012). Overall, these changes enhance current flow that underlies synaptic transmission at central synapses. In contrast, another core auxiliary protein, GSG1L (germ cell-specific gene 1-like protein), not only increases CP-AMPAR sensitivity to intracellular polyamines but also greatly decreases their single-channel conductance (McGee et al., 2015). Thus, despite both GSG1L and TARPs being claudin superfamily proteins with a common topology (Price et al., 2005; Shanks et al., 2012; Ramos-Vicente et al., 2021), they have directly opposite effects on AMPAR function and synaptic transmission (McGee et al., 2015; Gu et al., 2016). Although the novel behavior of GSG1L is striking, neither the mechanism by which it suppresses current through CP-AMPARs nor the role played by intracellular polyamines in this process is well understood.

Here we investigated the interplay between GSG1L and intracellular spermine in the regulation of CP-AMPARs. We find that, in marked contrast with action of canonical AMPAR auxiliary subunits, GSG1L's behavior is unique in that intracellular polyamines are an essential requirement, enabling GSG1L to trigger key changes that negatively regulate CP-AMPAR properties.

Materials and Methods

Heterologous expression. Recombinant AMPAR subunits, TARPs, and GSG1L were expressed in human embryonic kidney (HEK) 293 cells, a gift from T. Smart (University College London). We used rat *Gria2*, Q/R, and R/G edited flip form (pcDNA3.1-GluA2), originally from P. Seeburg. Rat *Gsg1l* (pcDNA3.1-GSG1L), courtesy of B. Fakler (University Freiburg), and rat *Cacng2* (pIRES-eGFP-γ2), courtesy of R. Nicoll (University of California, San Francisco), were transfected in excess of AMPAR subunits (2:1) to ensure coassembly. Transient transfection was performed with Lipofectamine 2000 (Invitrogen) according to the directions of the manufacturer. In all transfections, the total amount of DNA was 0.8 μg. Cells were split 12–24 h after transfection and plated on glass coverslips. Electrophysiological recordings were performed 24–48 h later.

AMPAR subunit and GSG1L mutagenesis. Based on the predicted transmembrane domains of GSG1L (Uniprot), the C-terminal domain (94 amino acids total) begins at amino acid position N229 as follows: NSYTKTVIEFRHKRKVF. In $GSG1L_{1\text{-}228}$, the sequence for the C-terminal domain was deleted in its entirety, while in $GSG1L_{1\text{-}238}$, a stop codon was inserted after F238, deleting 84 residues but retaining the first 10 membrane-proximal amino acids of the C-terminal domain. These shorter forms of GSG1L and the GluA2 point mutations C610S and D611N at positions +3 and +4 from the Q/R site were generated by performing PCR on pcDNA3.1-GSG1L and pcDNA3.1-GluA2, respectively, using the phosphorylated primers found in Table 1.

Fast agonist application to excised patches. We recorded macroscopic currents from outside-out patches using an Axopatch 700B amplifier, as previously described (Soto et al., 2007). Records were low-pass filtered at 6 kHz and digitized at 20 kHz using either the WINWCP software (Strathclyde Electrophysiology Software; John Dempster, University of Strathclyde) or pClamp 10 software (Molecular Devices) and analyzed using Igor Pro 8.04 (WaveMetrics) with NeuroMatic 3.0 (Rothman and Silver, 2018). Pipettes were pulled from thick-walled borosilicate glass (1.5 mm outer diameter, 0.86 mm inner diameter; Harvard Apparatus) and fire polished to a final resistance of ∼6–8 MΩ. The "external" solution contained the following (in mM): 145 NaCl, 2.5 KCl, 1 $CaCl_2$, 1 $MgCl_2$, 10 glucose, and 10 HEPES buffer, pH 7.3, with NaOH. Rapid application of glutamate (10 mM) was achieved by piezoelectric translation of a theta-barrel application tool. The "internal" (pipette) solution contained the following (in mM): 145 CsCl, 2.5 NaCl, 1 Cs-EGTA, 4 Mg-ATP, and 10 HEPES, pH 7.3, with CsOH and 0.1 to 1 mM spermine tetrahydrochloride (Tocris Bioscience), as indicated.

In some recordings, spermine was omitted, and endogenous polyamines were chelated by adding 20 mM Na_2ATP to the intracellular solution (Watanabe et al., 1991). In these cases, chelation/washout of endogenous polyamines was monitored by examining the gradual loss of inward rectification, assessed as changes in the rectification index (calculated as ratio of peak currents at +80 and −80 mV; $RI_{+80/-80}$). For GluA2(Q) alone, washout was rapid: the initial $RI_{+80/-80}$ value, measured soon after patch excision, was 1.58 ± 0.2 ($n = 7$), indicating outward rectification, a characteristic of CP-AMPARs in the absence of polyamines (McGee et al., 2015). When GluA2(Q) was expressed with GSG1L, washout was markedly slower. Responses were initially inwardly rectifying ($RI_{+80/-80}$ 0.49 ± 0.08; $n = 6$), and inward rectification was largely eliminated only after 10 min (0.83 ± 0.13; $n = 5$). At 20 min, outward rectification was observed (1.25 ± 0.13; $n = 3$). For experiments reporting the effects of polyamine chelation/washout, all measurements were obtained >10 min after patch excision.

To generate conductance–voltage (G–V) curves, we converted the peak currents to conductances based on the holding potential. To account for the polyamine-independent outward rectification of AMPARs, we divided the conductance values by those obtained in the polyamine-free condition. For currents that displayed inward rectification only, G–V curves were fitted with the following Boltzmann equation:

$$G = G_{\max}\left(\frac{1}{1 + \exp\left(\frac{V_{\mathrm{m}} - V_{\mathrm{b}}}{k_{\mathrm{b}}}\right)}\right),$$

where $G_{\max}$ is the conductance at a sufficiently hyperpolarized potential to produce full relief of polyamine block, V_{m} is the membrane potential,

Table 1. List of primers used

Construct	Primer name	Primer sequence
GluA2(Q)-D611N	GluA2 D611N Fwd	AATATTTCGCCAAGATCCCTCT
	GluA2 D611N Rev	GCATCCTTGTTGCATAAAGGC
GluA2(Q)-C610S	GluA2 C610S Fwd	TCCGATATCTCGCCAAGATCC
	GluA2 C610S Rev	TCCCTGCTGCATAAAGGC
$GSG1L_{1\text{-}228}$	$GSG1L_{1\text{-}228}$ Fwd	CTCTGACTCGAGTCTAGAGGGCCCGTTTA
	$GSG1L_{1\text{-}228}$ Rev	CGTAGTGACGGACGCCGCCATGCAGCAGGT
$GSG1L_{1\text{-}238}$	$GSG1L_{1\text{-}238}$ Fwd	TCCTATACCAAGACAGTCATT
	$GSG1L_{1\text{-}238}$ Rev	TTAGAGCGTAGTGACGGACG

V_b is the potential at which 50% of block occurs, and k_b is a slope factor describing the voltage dependence of block (the membrane potential shift necessary to cause an e-fold change in conductance). For currents that displayed double rectification, G–V curves were fitted with a double Boltzmann equation which contains equivalent terms for voltage-dependent permeation (p; Panchenko et al., 1999) as follows:

$$G = G_{\max}\left(\frac{1}{1+\exp\left(\frac{V_m - V_b}{k_b}\right)}\right) + G_{\max,\,p}\left(\frac{1}{1+\exp\left(\frac{V_m - V_p}{-k_p}\right)}\right),$$

To extract channel properties from macroscopic responses, we used nonstationary fluctuation analysis (NSFA) routines written in Igor Pro, as described previously (Soto et al., 2007). The weighted-mean single-channel current (i) and the total number of channels (N) were determined by plotting the ensemble variance (σ^2) against mean current ($\bar{I}$) and fitting with the following equation:

$$\sigma^2 = \frac{i\bar{I} - \bar{I}^2}{N + \sigma_B^2},$$

where $\sigma^2{}_B$ is the baseline variance.

Igor Pro was used to analyze channel openings in the tail of macroscopic patch currents (filtered at 1 kHz). Clear single-channel events (lasting longer than 2 ms) were selected by eye. For each event, a section of the baseline of equivalent length was also selected. An all-point amplitude histogram was generated and fitted with two Gaussians to determine the amplitude of the single-channel current. On average, 59 channel events (range, 36–69) were measured from each patch. Current–voltage (I–V) relationships were examined from −80 to +100 mV, and rectification index ($RI_{+80/-80}$) values were calculated as the ratio of the peak current at +80 and −80 mV. Recovery from steady-state desensitization was examined using a double-pulse protocol at −60 mV, consisting of a pair of 100 and 10 ms glutamate applications separated by increasing intervals. Peak currents recorded in response to the second application were normalized to the maximal current recorded during the first glutamate application and data points fit to mono- or biexponential functions to characterize $\tau_{w,rec}$ (ms).

Experimental design and statistical analysis. Summary data are presented in the text as mean ± SEM from n cells. Comparisons involving two datasets only were performed using an unpaired two-sided Welch two-sample t test that does not assume equal variance. Data from three or more groups were tested using two-way analysis of variance (ANOVA; Welch heteroscedastic F test) followed by pairwise comparisons using two-sided Welch two-sample t tests with Holm's sequential Bonferroni's correction for multiple comparisons. Estimates of unpaired mean differences and their bias-corrected and accelerated 95% confidence intervals from bootstrap resampling are presented as effect size (lower bound, upper bound). Exact p values are presented to two significant places, except where $p < 0.0001$. Differences were considered significant at $p < 0.05$. Statistical tests were performed using R (version 4.3.1, the R Foundation for Statistical Computing, http://www.r-project.org/) and R Studio (version 2023.12.1, Posit Software). No statistical test was used to predetermine sample sizes; these were based on standards of the field. No blinding or randomization was used.

Results

GSG1L and TARP γ2 have contrasting effects on CP-AMPAR spermine block

To directly compare the influence of GSG1L and the prototypical auxiliary subunit TARP γ2 on the voltage-dependent behavior of CP-AMPARs, we recorded currents evoked by rapid application of glutamate (10 mM, 100 ms) to outside-out membrane patches from transfected HEK cells held between −100 and +80 mV (Fig. 1*A*). In the presence of added intracellular spermine (0.1 mM), homomeric GluA2(Q) CP-AMPARs displayed inwardly rectifying current–voltage (I–V) relationships (Fig. 1*B*), with a rectification index ($RI_{+80/-80}$) of 0.17 ± 0.08 (mean ± SEM; $n = 18$). The inward rectification was markedly increased by coexpression of GSG1L [$RI_{+80/-80}$ 0.03 ± 0.007; $n = 10$; $t_{(20.94)} = 6.96$; $p < 0.0001$ compared with GluA2(Q) alone] and reduced by coexpression of γ2 ($RI_{+80/-80}$ 0.54 ± 0.12; $n = 9$; $t_{(11.81)} = -8.69$; $p < 0.0001$; Fig. 1*B*,*C*).

Next, to investigate the influence of spermine concentration, we made recordings with 0.5 and 1.0 mM added intracellular spermine. Comparison of conductance–voltage (G-V) relationships (Fig. 1*D*) showed that the differential effects of GSG1L and γ2 on outward current flow at positive potentials were maintained in elevated spermine. Additionally, fits of the inwardly or doubly rectifying G–V curves with single or double Boltzmann functions (Panchenko et al., 1999) showed that as the concentration of added spermine was increased, there was a progressive negative shift in V_b (the potential at which 50% block occurs; Fig. 1*E*). Two-way ANOVA showed main effects on V_b of both auxiliary subunit (none, GSG1L or γ2) and spermine concentration (0.1, 0.5, or 1 mM) and an interaction between auxiliary subunit and spermine concentration (auxiliary, $F_{(2,56)} = 724.08$; $p < 0.0001$; spermine, $F_{(2,56)} = 155.88$; $p < 0.0001$; interaction, $F_{(4,56)} = 9.01$; $p < 0.0001$; $n = 3$–16). Although not observed in our previous study (McGee et al., 2015), our experiments identified an effect of GSG1L on the voltage dependency of spermine block at negative potentials. The present findings indicate a real and sizeable difference in V_b between GluA2 and GSG1L (Table 2), while the previous study may have been underpowered to reliably detect a difference. Additionally, a plot of the unpaired mean difference between V_b with GluA2(Q) alone and that with either GSG1L or γ2 (ΔV_b) showed a differential effect of the two auxiliary subunits at all spermine concentrations (Fig. 1*E*; Table 2).

Spermine attenuation of channel conductance is enhanced by GSG1L

We found previously that GSG1L, unlike other auxiliary AMPAR subunits, reduced CP-AMPAR single-channel currents in the presence of intracellular spermine (McGee et al., 2015). To determine whether spermine is required for this attenuation, we examined GluA2(Q) receptors with and without GSG1L over a range of spermine concentrations (0, 0.1, 0.5, and 1.0 mM). Weighted-mean single-channel conductance was estimated using NSFA of currents evoked by 100 ms applications of 10 mM glutamate to outside-out patches (−60 mV; Fig. 2*A*). For comparison, we also examined the effect of the prototypical TARP γ2, which we anticipated would increase channel conductance regardless of the presence of intracellular spermine.

Overall, two-way ANOVA indicated main effects on GluA2(Q) single-channel conductance of auxiliary subunit type (none, GSG1L, γ2; $F_{(2,96)} = 130.70$; $p < 0.0001$) and spermine concentration ($F_{(3,96)} = 39.83$; $p < 0.0001$). The interaction approached, but did not reach, statistical significance ($F_{(6,96)} = 1.85$; $p = 0.098$), suggesting that the combined effect of these two factors on single-channel conductance may not be entirely uniform across different levels of the factors. Pairwise comparisons confirmed that in the presence of 0.1 mM spermine, while TARP γ2 increased the conductance of GluA2(Q) receptors by ∼50% (from 20.6 to 30.9 pS), GSG1L reduced the conductance by ∼40% (to 12.7 pS) (Fig. 2*B*; Table 3). Second, we found that, in the absence of auxiliary subunits, increasing the intracellular concentration of spermine reduced single-channel conductance

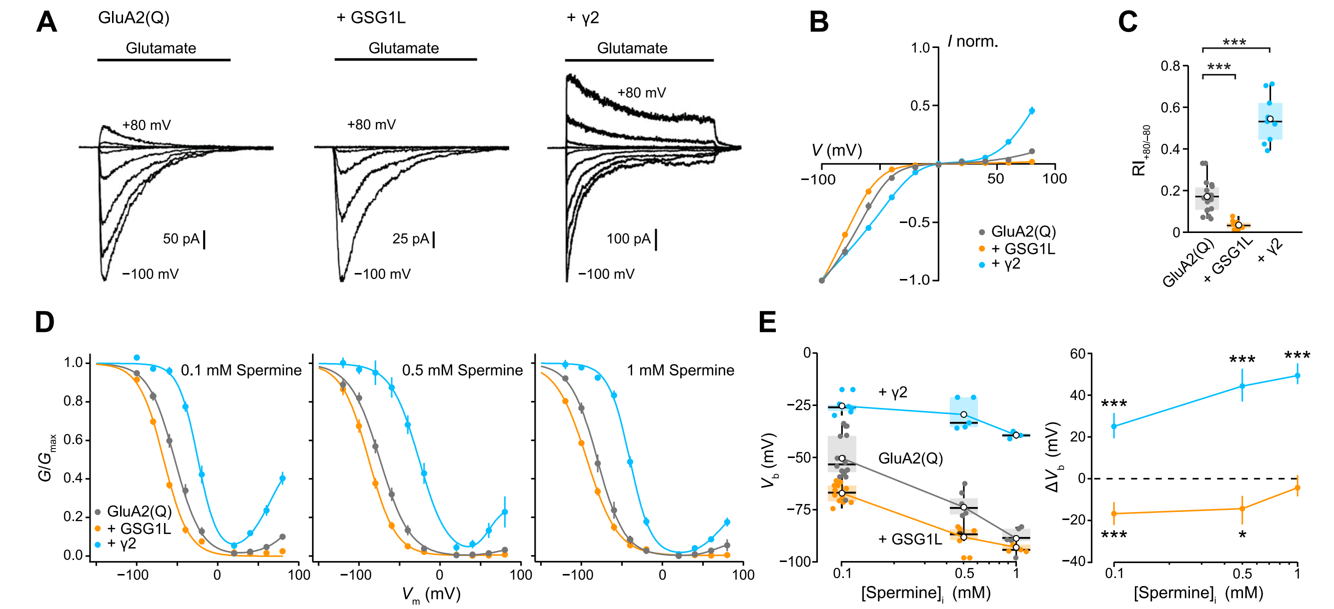

Figure 1. GSG1L and TARP γ2 differentially modulate spermine block of CP-AMPAR channels. ***A***, Glutamate-evoked currents from homomeric GluA2(Q), in the absence and presence of auxiliary subunits GSG1L and γ2, at holding potentials between −100 and +80 mV (Δ20 mV). Currents were activated by 100 ms applications of 10 mM glutamate to outside-out patches from transfected HEK cells (with 0.1 mM intracellular spermine). ***B***, *I–V* relationships constructed for the peak current responses shown in ***A***. Error bars denote SEM, and curves are weighted fits of seventh-order polynomials. ***C***, Pooled $RI_{+80/-80}$ data. Box-and-whisker plots indicate the median (black line), the 25–75th percentiles (box), and the 10–90th percentiles (whiskers); filled circles are data from individual patches and open circles indicate means. $***p < 0.001$ (unpaired Welch two-sample *t* tests). ***D***, Plots of normalized conductance against voltage for GluA2(Q), in the absence and presence of GSG1L or γ2, with different intracellular spermine concentrations (0.1, 0.5, and 1 mM). Lines are fits of single or double Boltzmann functions, and vertical error bars indicate SEM. ***E***, Left, pooled V_b values (the potential at which 50% block of inward current occurred) from Boltzmann fits to *G–V* plots for individual patches. Box-and-whisker plots as in ***C***. Right, Plot showing the shift in V_b values produced by GSG1L and γ2. Symbols indicate unpaired mean differences from GluA2(Q) alone (ΔV_b), and error bars denote 95% CI. $*p < 0.05$; $***p < 0.001$ (unpaired Welch two-sample *t* tests).

Table 2. Effect of GSG1L and γ2 on the voltage dependence of GluA2(Q) block by intracellular spermine

			ΔV_b compared with GluA2(Q) alone			
$[\text{Spermine}]_i$ (mM)	Receptor	V_b (mV)	upMD [95% CI]	*t*	df	*p*
0.1	GluA2(Q)	−50.5 ± 2.7 (16)				
	+GSG1L	−67.2 ± 1.3 (12)	−16.7 [11.3, 22.4]	−5.64	20.90	<0.0001
	+γ2	−25.5 ± 1.8 (6)	25.1 [−31.0, −19.1]	7.82	19.75	<0.0001
0.5	GluA2(Q)	−73.9 ± 2.9 (7)				
	+GSG1L	−88.2 ± 2.2 (6)	−14.3 [7.7, 21.1]	−3.89	10.70	0.0027
	+γ2	−29.5 ± 3.4 (5)	44.4 [−52.7, −36.8]	9.93	8.89	<0.0001
1.0	GluA2(Q)	−88.9 ± 2.6 (5)				
	+GSG1L	−93.1 ± 1.1 (5)	−4.3 [−1.7, 8.5]	−1.50	5.33	0.19
	+γ2	−39.4 ± 1.0 (3)	49.4 [−55.7, −45.4]	17.42	5.06	<0.0001

Summary of V_b values presented as mean ± SEM, from (*n*) patches. The effects of GSG1L and γ2 are shown as unpaired mean differences (upMD) compared with GluA2(Q) alone, with 95% confidence intervals [lower bound, upper bound]. Pairwise tests against GluA2(Q) are unpaired two-sided Welch *t* tests reporting the test statistic (*t*), degrees of freedom (df), and *p* values adjusted for multiple comparisons using Holm's sequential Bonferroni's correction.

estimates for GluA2(Q) in a concentration-dependent manner. While the conductance was similar with 0 and 0.1 mM spermine (21.0 and 20.6 pS, respectively), increasing spermine to 0.5 or 1.0 mM reduced the conductance by ∼50% (to 10.6 and 9.1 pS, respectively; Fig. 2*B*,*C*; Table 3).

Coexpression of TARP γ2, in the absence of added spermine, increased the estimated single-channel conductance of GluA2(Q) to 32.5 pS, but there was no concentration-dependent reduction in conductance as spermine was increased (Fig. 2*B*,*C*; Table 3). In contrast, when GSG1L was cotransfected with GluA2(Q), increasing intracellular spermine decreased channel conductance. With 0.5 and 1.0 mM added spermine, the conductance was reduced by ∼60% (Fig. 2*B*; Table 3). Of note, in the absence of spermine the conductance with GSG1L (17.4 ± 1.2 pS; $n = 11$) was not different from that of GluA2(Q) alone (21.0 ± 1.9; $n = 11$; $t_{(17.11)} = -1.57$; $p_{adj} = 0.67$), suggesting that the reduction in conductance with GSG1L required the presence intracellular spermine. Consistent with this idea, in the absence of spermine, GSG1L also failed to alter the single-channel conductance of a second CP-AMPAR, homomeric GluA4 (21.8 ± 1.0 pS for GluA4 and 19.6 ± 1.0 pS with GSG1L; $n = 5$ and 7, respectively; $t_{(9.72)} = 1.47$; $p = 0.17$). This contrasts with the GSG1L-induced reduction in the conductance of GluA4 homomers seen in the presence of 0.1 mM intracellular spermine (McGee et al., 2015). Moreover, in the presence of 0.1 mM intracellular spermine, GSG1L coexpression had no effect on the conductance of spermine-insensitive CI-AMPARs composed of GluA2(R) (2.7 ± 0.3 pS alone vs 3.4 ± 0.3 pS with GSG1L; $n = 5$

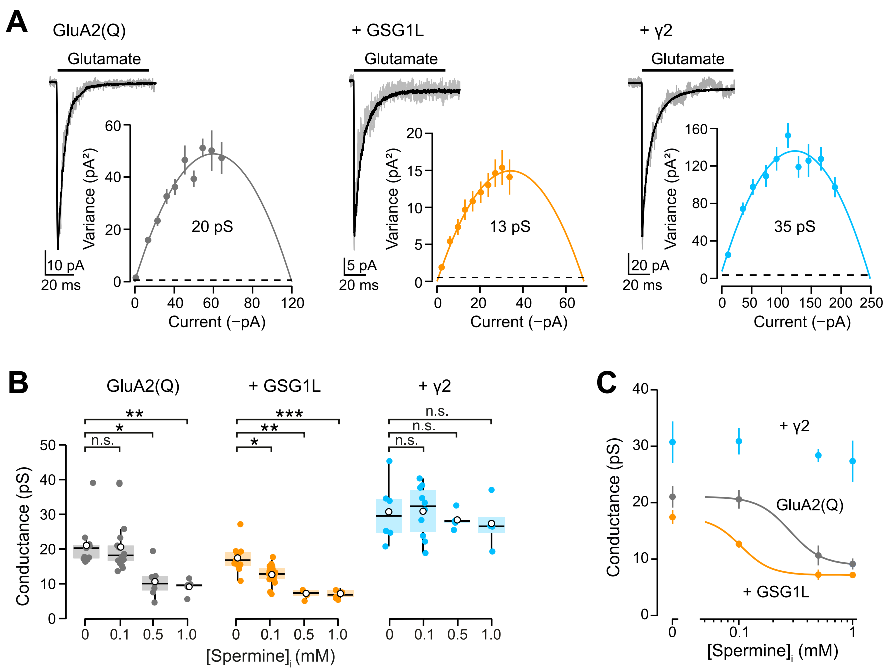

Figure 2. Spermine and GSG1L reduce estimated single-channel conductance of CP-AMPARs. ***A***, Representative individual and averaged currents (gray and black, respectively) evoked by glutamate (10 mM, 100 ms) application to patches (−60 mV) from cells expressing GluA2(Q) alone (left), GluA2(Q) + GSG1L (middle), or GluA2(Q) + γ2 (right), together with the corresponding current–variance relationship from NSFA of the desensitizing current phase, yielding the indicated weighted-mean single-channel conductance estimates. Symbols indicate mean, and error bars are SEM. Dashed lines denote baseline variance. ***B***, Pooled conductance estimates obtained under different conditions with different spermine concentrations. Box-and-whisker plots as in Figure 1*C* (*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; Table 3). ***C***, Data from ***B*** plotted as concentration–response curves. Solid lines are from a simultaneous fit of the two conditions to the Hill equation with a common Hill slope. For GluA2(Q) and GSG1L, where spermine caused significant reductions in conductance (as shown in ***B***), the EC_{50} estimates were 282 and 104 μμ, respectively. Symbols indicate mean, and error bars are SEM.

and 11, respectively; unpaired mean difference, 0.7 pS [0.02, 1.5]; $t_{(10.92)} = 1.70$; $p = 0.12$), GluA1/2(R) (3.7 ± 0.3 pS vs 3.4 ± 0.6 pS with GSG1L; $n = 10$ and 7, respectively; unpaired mean difference −0.2 pS [−1.3, 1.1]; $t_{(9.34)} = 0.34$; $p = 0.74$), or GluA4/2(R) (3.1 ± 0.2 pS vs 3.1 ± 0.3 pS with GSG1L; $n = 5$; unpaired mean difference 0.0 pS [−0.8, 0.5]; $t_{(5.6)} = -0.07$; $p = 0.94$).

Plotting the GluA2(Q) conductance data as concentration–response curves (Fig. 2*C*) suggested that incorporation of GSG1L increased the potency of spermine at GluA2(Q) receptors, with the EC_{50} reduced from ∼280 μM to ∼100 μM. Taken together, these findings support the view that intracellular spermine selectively decreases the conductance of CP-AMPARs and that coassembly with GSG1L enhances this effect while coassembly with TARP γ2 nullifies it.

Spermine attenuates the apparent conductance of directly resolved GluA2(Q)/GSG1L channels

To examine more directly the effect of spermine on single-channel conductance, we analyzed resolved GluA2(Q)/GSG1L channel openings in the tail of glutamate-evoked macroscopic currents (at −120 mV). We determined the amplitude of selected individual events from Gaussian fits to all-point histograms (Fig. 3*A*). The inclusion of 1 mM intracellular spermine reduced mean channel amplitudes by ∼30% (from 21.2 ± 1.4 pS to 15.7 ± 0.3 pS; $n = 5$; 57–69 and 37–68 events per patch, respectively; $t_{(4.37)} = 3.82$; $p = 0.016$; Fig. 3*B*,*C*). These results confirm that in addition to producing a voltage-dependent open–channel block (clearly apparent from the virtual absence of macroscopic current at positive potentials; Fig. 1*A*), spermine also diminishes current flow through individual GSG1L-associated CP-AMPARs channels at negative potentials.

Previous work has shown that AMPARs exhibit multiple conductance states (Swanson et al., 1997; Rouach et al., 2005; Tomita et al., 2005; Coombs and Cull-Candy, 2021). While TARPs have been reported to act either to increase the relative proportion of higher conductance substates (Tomita et al., 2005) or to increase the absolute conductance of all individual substates (Shelley et al., 2012), corresponding information is not available for GSG1L. In the present study, channel openings were recorded in response to 100 ms applications of 10 mM glutamate. Patches were first held at −60 mV to examine macroscopic desensitization kinetics and then switched to −120 mV and filtered at 1 kHz to increase the signal-to-noise ratio of channels in the current tail. The RMS noise in these conditions was relatively high (0.21–0.35 pA^2), and all-point amplitude histograms were used to determine mean current amplitudes. We were therefore not able to distinguish between changes in the proportion of the conductance substates or changes in the amplitude of all individual substates.

We next considered whether the reduction in estimated mean single-channel conductance caused by GSG1L and spermine could, in part, reflect an increase in fast "flickery" channel block by polyamines, as described for other examples of ion channel block (Neher and Steinbach, 1978; Yellen, 1984). For example, if spermine acts as a weak low-affinity channel blocker of CP-AMPARs and hence shows rapid dissociation kinetics, the resultant fast current interruptions may be subject to low-pass filtering due to the temporal resolution of the recording system leading to an underestimated single-channel conductance (Yang and Sigworth, 1998; Pusch et al., 2000). Although we cannot exclude this possibility, we think this is unlikely as the reduction in estimated conductance produced by GSG1L (∼40% with 0.1 mM spermine) was maintained across a range of filtering

Table 3. Effect of GSG1L and γ2 on GluA2(Q) conductance estimates from NSFA

			Compared with 0 mM spermine			
	$[\text{Spermine}]_i$ (mM)	Conductance (pS)	upMD [95% CI]	t	df	p
GluA2(Q)	0	21.0 ± 1.9 (11)				
	0.1	20.6 ± 7.1 (19)	−0.5 [−6.5, 3.4]	0.18	23.03	1.00
	0.5	10.6 ± 1.8 (7)	−10.4 [−15.8, −6.1]	3.98	15.55	0.010
	1.0	9.1 ± 1.0 (5)	−11.9 [−17.5, −8.9]	5.52	13.58	0.00093
GluA2(Q) + GSG1L	0	17.4 ± 1.2 (11)				
	0.1	12.7 ± 0.6 (18)	−4.8 [−7.9, −2.6]	3.43	15.28	0.025
	0.5	7.2 ± 0.9 (3)	−10.2 [−13.1, −7.7]	6.63	9.70	0.00082
	1.0	7.2 ± 1.2 (5)	−10.3 [−13.2, −8.0]	7.61	13.18	<0.0001
GluA2(Q) + γ2	0	32.5 ± 2.9 (10)				
	0.1	30.9 ± 2.3 (10)	−1.6 [−9.4, 4.6]	−6.26	9.49	1.00
	0.5	28.4 ± 1.1 (5)	−4.1 [−10.3, 1.2]	−2.77	10.97	0.89
	1.0	27.4 ± 3.7 (4)	−3.4 [−13.3, 3.2]	−5.89	16.16	0.92

Summary of weighted-mean single-channel conductance presented as mean ± SEM, from (n) patches. The effects of different added spermine concentrations are shown as unpaired mean differences (upMD) compared with 0 spermine, with 95% confidence intervals [lower bound, upper bound]. Pairwise tests (against 0 spermine) are unpaired two-sided Welch t tests reporting the test statistic (t), degrees of freedom (df), and p values adjusted for multiple comparisons using Holm's sequential Bonferroni's correction. For clarity, other pairwise tests comparing across receptors (performed as part of the same family of tests) are reported in the text.

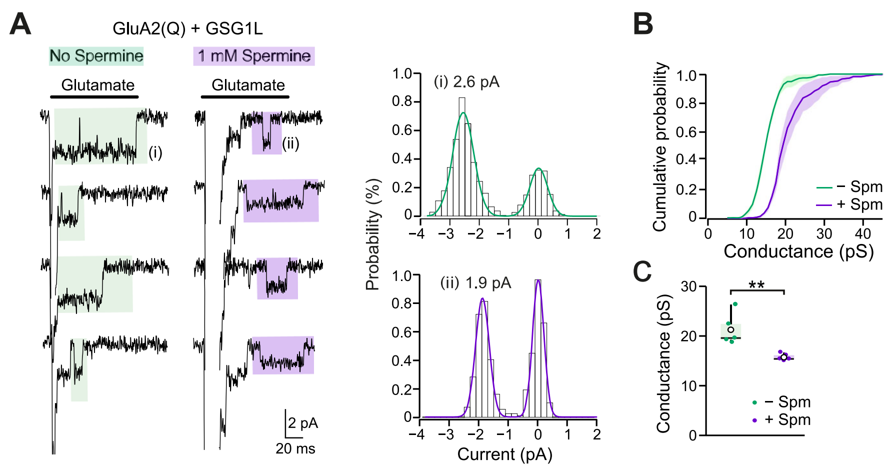

Figure 3. Single-channel conductance of directly resolved GluA2Q/GSG1L openings is reduced by intracellular spermine. ***A***, Representative single-channel currents activated by application of glutamate (10 mM, 100 ms) to outside-out patches (−60 mV) from cells expressing GluA2(Q) and GSG1L, with and without 1 mM intracellular spermine. Single-channel openings present in the tail of truncated macroscopic currents (1 kHz low-pass filter) are highlighted by colored boxes. To the right are all-point amplitude histograms of the indicated individual channel openings (***i*** and ***ii***) and the calculated amplitudes of these openings. ***B***, Averaged cumulative probability distributions of single-channel events from five patches (36–69 per patch) in each condition (fills denote SEM). (**C**) Box-and-whisker plots (as in Fig. 1*C*) illustrating the effect of 1 mM intracellular spermine on mean single-channel conductance from five patches in each condition (**$p < 0.01$; unpaired Welch two-sample t test).

frequencies (0.1–5 kHz range; data not shown). Moreover, open-channel block is expected to be near complete at high concentrations of the blocking molecule (Green and Andersen, 1991). Our results indicate that for homomeric GluA2(Q) both with and without GSG1L, the spermine-induced reduction in apparent channel conductance plateaus at ∼50% (Fig. 2*B*,*C*), suggesting a simple open-channel blocking mechanism is inconsistent with these effects.

D611 is required for GSG1L- and spermine-mediated attenuation of GluA2 conductance

Calcium permeability, high channel conductance, and polyamine block all depend critically on the presence of a neutral glutamine residue at the Q/R site in the AMPAR channel's ion selectivity filter (Burnashev et al., 1992; Bowie and Mayer, 1995; Kamboj et al., 1995; Swanson et al., 1997). It has also been shown that the neighboring conserved negatively charged aspartate residue at the cytoplasmic entrance of the channel pore (Q/R+4 position) is critical for polyamine block (Panchenko et al., 1999). As both polyamine block and single-channel conductance are altered by GSG1L and because the equivalent +4 site in GluA1 has been shown to be crucial for TARP enhancement of channel conductance (Soto et al., 2014), we next asked whether the presence of GSG1L modifies the influence of this residue (D611 in GluA2) on ion permeation.

When we replaced the negatively charged aspartate 611 in GluA2(Q) with a neutral asparagine (D611N), we obtained a near-linear *I–V* relationship (Fig. 4*A*), indicating a reduced influence of spermine as previously observed with the same mutation at the +4 site in GluA1 (Soto et al., 2014). Inclusion of GSG1L enhanced spermine block of the D611N mutant, reducing $RI_{+80/-80}$ from 1.12 ± 0.03 to 0.38 ± 0.07 ($n = 9$ and 4, respectively; $t_{(3.78)} = 9.48$; $p = 0.00091$; Welch t test; Fig. 4*A*,*B*). For the D611N mutant, the single-channel conductance was 4.5 ± 0.3 pS ($n = 9$), and coexpression of GSG1L did not decrease this (6.2 ± 0.8 pS; $n = 4$; $t_{(3.84)} = 2.06$; $p = 0.22$; Welch t test; Fig. 4*C*). Moreover,

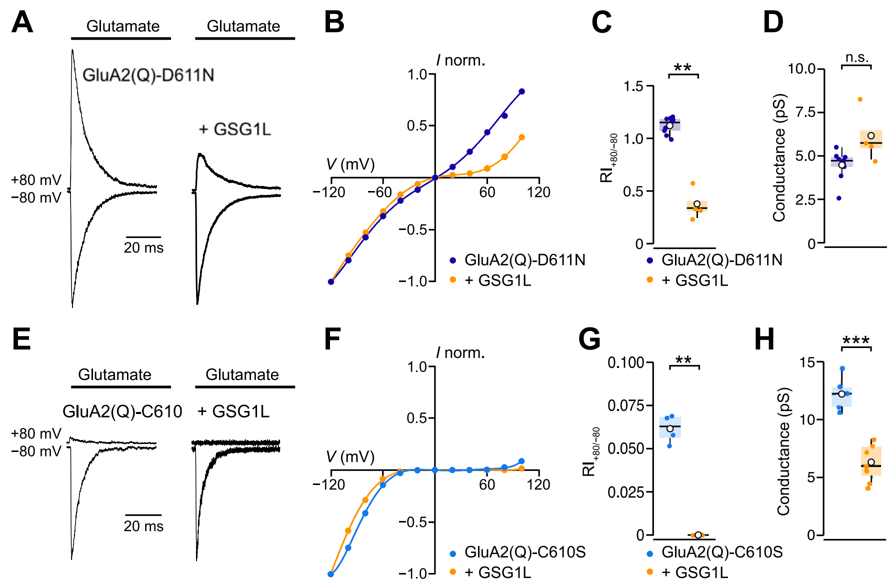

Figure 4. The aspartate residue at Q/R + 4 in GluA2 is not necessary for GSG1L to enhance polyamine block but is required for the decrease in single-channel conductance. ***A***, Representative glutamate-evoked currents (10 mM; 100 ms; recorded at +80 and −80 mV) from GluA2(Q)-D611N, in the absence and presence of GSG1L (0.1 mM spermine). ***B***, Normalized peak *I*–*V* relationships (global average; $n = 9$ and 4), showing increased rectification at positive potentials in the presence of GSG1L (100 µM spermine). Error bars denote SEM, and curves are weighted fits of seventh-order polynomials. ***C***, Pooled RI data. ***D***, Box-and-whisker plots showing pooled NSFA data (−60 mV) for GluA2(Q)-D611N, with GSG1L and in the presence of 1 mM intracellular spermine. Note GluA2(Q)-D611N is no longer sensitive to GSG1L-induced or spermine-induced reduction in conductance. ***E–H***, As in ***A–D*** but for GluA2(Q)-C610S. Although the peak currents for GluA2(Q)-C610S at +80 mV (in the absence of GSG1L) were small (11.4 ± 5.0 pA; $n = 5$), they were clearly measurable [21.1 (range, 11.9–43.9) times greater than the baseline SD at 2 kHz filtering]. With GSG1L no outward currents were detected at +80 mV. Note that for GluA2(Q)-C610S, GSG1L increased rectification and reduced estimated single-channel conductance.

single-channel conductance of GluA2(Q)-D611N was not attenuated when intracellular spermine was increased from 0.1 to 1 mM (4.4 ± 0.4 pS; $n = 5$; $t_{(8.20)} = -0.19$; $p = 0.86$; Welch *t* test), raising the possibility that Asp 611 is crucial for GSG1L's spermine-dependent influence on channel function.

We also examined the impact of mutating a neighboring residue in the ion permeation pathway, replacing the cysteine residue at the Q/R + 3 site of GluA2(Q) with a serine (C610S). In this case, GSG1L both increased rectification (reduced $RI_{+80/-80}$ from 0.062 ± 0.004 to 0.0 ± 0.0; $n = 4$ and 5; $t_{(3.0)} = 14.65$; $p = 0.0007$; Fig. 4*D*,*E*) and decreased mean channel conductance (from 11.6 ± 0.8 pS to 6.0 ± 0.6 pS; both $n = 6$ and 6; $t_{(8.91)} = 5.41$; $p = 0.00044$; Welch *t* tests; Fig. 4*F*). Together these results suggest that the Q/R + 4 site aspartate residue (but not the +3 cysteine residue) is a key determinant in GSG1L's ability to decrease CP-AMPAR channel conductance and further implicate spermine in this effect.

The proximal C-tail region of GSG1L affects the voltage dependency of spermine block at negative potentials

Our earlier work indicated that deleting the C-tail of γ2 modifies the effect of TARPs on polyamine block and channel conductance (Soto et al., 2014). We therefore next asked whether the C-tail of GSG1L was required for its influence on CP-AMPARs, by either deleting the entire C-tail ($GSG1L_{1-228}$) or truncating it, sparing the 10 membrane-proximal amino acids ($GSG1L_{1-238}$).

We found that both mutants retained the ability to increase GluA2(Q) inward rectification [$RI_{+80/-80}$; $GSG1L_{1-228}$ 0.048 ± 0.006 and $GSG1L_{1-238}$ 0.044 ± 0.009; $t_{(21.0)} = 6.28$; $p < 0.0001$ and $t_{(23.0)} = 6.15$; $p < 0.0001$ vs WT GluA2(Q)]. However, while GSG1L and $GSG1L_{1-238}$ produced a leftward shift in the negative limb of the *G*–*V* relationship [unpaired mean difference from GluA2(Q) V_b −26.1 mV [−34.1, −19.3]; $t_{(11.07)} = 6.27$; $p = 0.00018$ and −18.6 mV [−27.6, −10.9]; $t_{(14.56)} = 4.04$; $p = 0.0023$, respectively; $n = 6$–10], $GSG1L_{1-228}$ did not (−4.8 mV [−15.0, 5.0]; $t_{(17.89)} = 0.89$; $p = 0.39$; $n = 8$ and 12). This suggests that the proximal region of the C-tail of GSG1L plays a role in modulating AMPAR polyamine block—at least for inwardly flowing currents.

GSG1L's C-tail influences recovery from desensitization

Previous studies have shown that GSG1L slows CP-AMPAR entry into, and recovery from, desensitization; among the auxiliary subunits, this slowing of recovery is unique to GSG1L (Schwenk et al., 2012; Shanks et al., 2012; Perozzo et al., 2023). We found that both $GSG1L_{1-228}$ and $GSG1L_{1-238}$ were as effective as full-length GSG1L in slowing GluA2(Q) entry into desensitization. The unpaired mean difference in $\tau_{w,des}$ was 3.61 [2.53, 4.59] ms for GSG1L ($t_{(12.72)} = 6.37$; $p < 0.0001$), 5.00 [2.45, 7.91] ms for $GSG1L_{1-228}$ ($t_{(7.72)} = 3.34$; $p = 0.011$), and 4.86 [3.58, 6.18] ms for $GSG1L_{1-238}$ ($t_{(18.56)} = 6.84$; $p < 0.0001$; $n = 8$–14; Fig. 5*A*,*B*). Although the truncation mutants showed greater variance in $\tau_{w,des}$ values compared with full-length GSG1L, the reason for this is unclear.

To determine the effects of the C-tail truncations on recovery from desensitization, we analyzed responses to a two-pulse protocol, consisting of a pair of 100 and 10 ms glutamate applications separated by increasing time intervals. As expected, GSG1L slowed the time constant of recovery from desensitization of GluA2(Q) by roughly 10-fold (from 19.0 ± 2.6 to 224.1 ± 34.0 ms; $n = 8$; $t_{(7.09)} = 6.02$; $p = 0.0011$). $GSG1L_{1-238}$ behaved like the full-length protein, increasing the mean recovery time constant to 189.2 ± 29.2 ms ($n = 9$; $t_{(8.13)} = 5.81$; $p = 0.0011$). However, $GSG1L_{1-228}$ slowed recovery only modestly ($\tau_{w,rec} = 29.9 \pm 3.3$ ms; $n = 13$; $t_{(18.93)} = 2.58$ $p = 0.019$; Fig. 5*C*–*E*). Hence, the proximal 10 amino acid residues in GSG's C-tail appear crucial in this slowing process. Together, our results suggest that while the membrane-proximal region (positions 228–238) does not play a role in controlling the

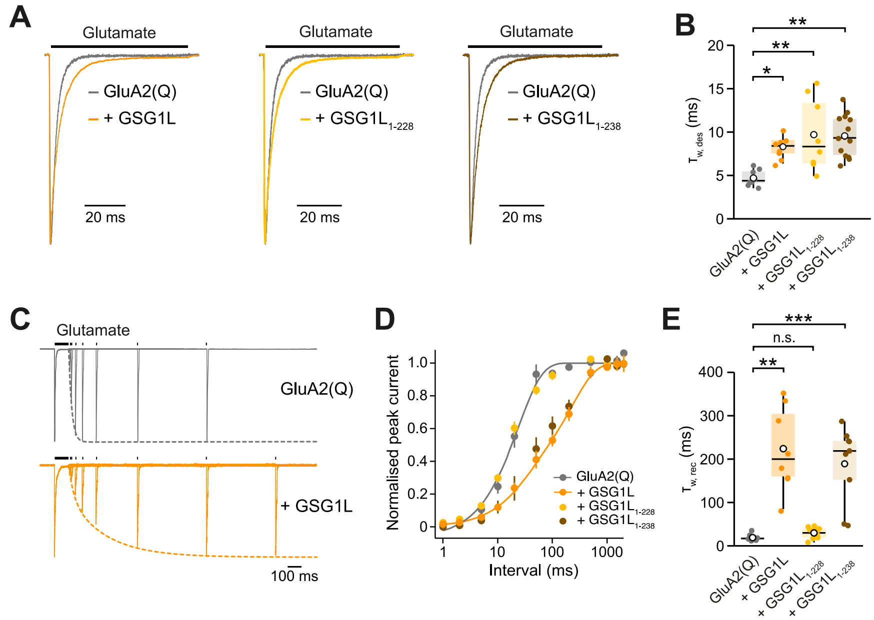

Figure 5. The proximal C-tail of GSG1L is needed for its effect on CP-AMPAR recovery from, but not entry into, desensitization. ***A***, Representative current responses (10 mM glutamate, 100 ms) from outside-out patches excised from HEK cells expressing GluA2(Q), either alone or in combination with full-length GSG1L or C-tail truncated GSG1L$_{1-228}$ or GSG1L$_{1-238}$ and 0.1 mM intracellular spermine. ***B***, Pooled data showing the time constants of desensitization ($\tau_{w,des}$) derived from biexponential fits to the current decline. Box-and-whisker plots as in Figure 1*C*. ***C***, Glutamate-evoked currents from cells expressing GluA2(Q), with or without GSG1L. A double-pulse protocol consisting of a pair of a 100 ms and a 10 ms glutamate pulses separated by increasing time intervals was used to determine the kinetics of recovery from desensitization. ***D***, Representative single patch recovery time courses for GluA2(Q) alone, with full-length GSG1L, with GSG1L$_{1-228}$, or with GSG1L$_{1-238}$. Data points are peak currents evoked by the second glutamate pulse normalized to the peak amplitude of the first glutamate application. Lines are mono- or biexponential fits. ***E***, Pooled $\tau_{w,rec}$ data. Box-and-whisker shows that GSG1L and GSG1L$_{1-238}$ increase GluA2(Q) recovery time constants by ∼10-fold, while GSG1L$_{1-228}$ has no effect.

kinetics of entry into desensitization, they are essential in GSG1L's ability to slow recovery from desensitization.

Spermine modulates the effect of GSG1L on recovery from desensitization

As the membrane-proximal C-tail of GSG1L appears able to influence both the action of intracellular spermine and recovery from desensitization, we asked whether the presence of intracellular spermine itself modifies GSG1L's influence on desensitization kinetics.

We found that for GluA2(Q) the slowing of entry into desensitization produced by GSG1L did not depend on the presence of intracellular spermine (Fig. 6*A*,*B*). Thus, GSG1L had similar effects regardless of whether 100 µM spermine was included in the intracellular solution or endogenous intracellular polyamines chelated by the addition of 20 mM Na_2ATP to the pipette solution (Watanabe et al., 1991). Two-way ANOVA showed a main effect of GSG1L on desensitization time constants but no main effect of spermine and no interaction (Table 4). A similar result was observed with GluA4 (Table 4). In contrast, the ability of GSG1L to slow recovery from desensitization was clearly influenced by intracellular spermine. Thus, GSG1L slowed the rate of recovery of GluA1, GluA2(Q), and GluA4 AMPARs in the presence of spermine, but this effect was diminished or eliminated when endogenous polyamines were chelated (Fig. 6*C*,*D*; Table 4). In each case, two-way ANOVA confirmed an interaction between the effects of GSG1L and spermine (Table 4). Surprisingly, therefore, spermine appears necessary for GSG1Ls ability to slow recovery from desensitization.

Discussion

We describe three main findings. First, intracellular spermine can markedly attenuate the conductance of CP-AMPAR channels. Second, this attenuation is enhanced when CP-AMPARs are coassembled with GSG1L but absent or greatly reduced when receptors are associated with TARP γ2. Third, slowing of CP-AMPAR recovery from desensitization by GSG1L is strongly influenced by the presence of intracellular spermine. Additionally, we find that the negatively charged residue D611 that lies close to the cytoplasmic entrance of the pore, is critical for the GSG1L/spermine modulation of single-channel conductance, while the proximal C-tail of GSG1L has a key role in the slowing of recovery from desensitization. Together, our findings reveal a novel and unexpected spermine dependence of the actions of GSG1L on CP-AMPARs.

Block by intracellular polyamines at depolarized potentials is a key property of CP-AMPARs (Bowie and Mayer, 1995; Donevan and Rogawski, 1995; Kamboj et al., 1995; Koh et al., 1995). Here, we show that intracellular spermine also decreases the single-channel conductance of auxiliary protein-lacking CP-AMPARs at negative voltages. Analogous dual roles of polyamines have been identified in inwardly rectifying (Kir2.1) potassium channels, where spermine was shown to reduce single-channel current amplitude without affecting open probability—an effect attributed to spermine screening of negatively charged acidic residues near the inner vestibule of the channel (Xie et al., 2002). Recent molecular dynamics simulations, modeling Kir2.2 channels, also suggest that weakly bound spermine can decrease ion permeation rates without influencing open probability (Jogini et al., 2023). Similarly, spermine (extracellular and intracellular) has been shown to reduce single-channel conductance of NMDA receptors through interaction with negatively charged pore residues (Rock and MacDonald, 1992, 1995; Araneda et al., 1999).

In calcium-permeable AMPA and kainate receptors, the electronegative selectivity filter provides the perfect environment for binding of positively charged polyamines (Twomey et al., 2019),

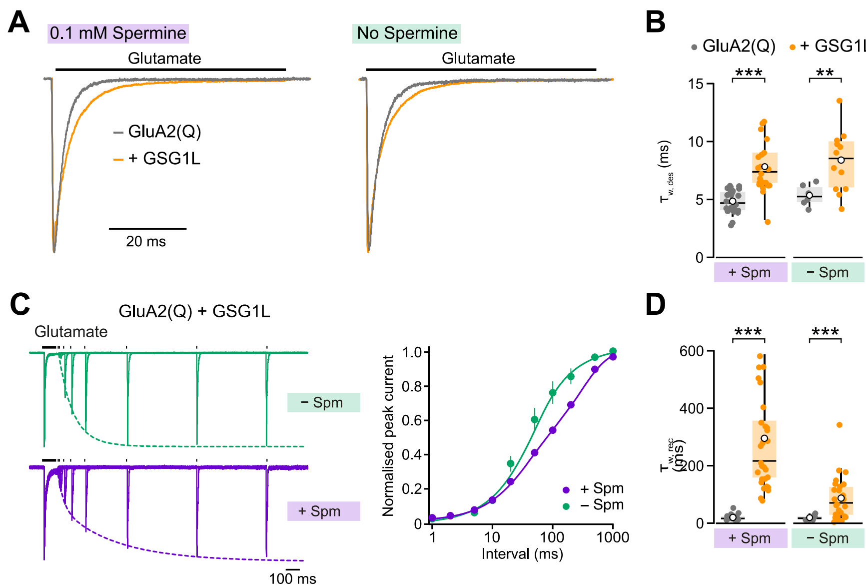

Figure 6. Spermine influences recovery from desensitization kinetics of GSG1L-containing CP-AMPARs. ***A***, Representative normalized glutamate-evoked currents (10 mM, 100 ms, −60 mV) from patches excised from HEK cells transfected with GluA2(Q) alone and GluA2(Q) + GSG1L, both with and without intracellular spermine (0.1 mM). ***B***, Box-and-whisker plots (as in Fig. 1C) showing pooled $\tau_{w,des}$ data. Note that GSG1L slows entry into desensitization irrespective of spermine concentration (Table 3). ***C***, Left, Representative currents illustrating protocols used to determine the kinetics of recovery from desensitization. Cells coexpressed GluA2(Q) and GSG1L and currents were recorded in the absence and presence of intracellular spermine. Right, Averaged recovery data. Data points peak currents evoked by the second glutamate application, normalized to the peak currents evoked by the first glutamate application. Lines are mono- or biexponential fits, and error bars denote SEM. ***D***, Box-and-whisker (as in Fig. 1C) showing pooled $\tau_{w,rec}$ data (Table 3).

and mutations in this region reduce spermine block (Panchenko et al., 1999; Panchenko et al., 2001; Wilding et al., 2010; Soto et al., 2014). D611, for example, which sits in the Q/R + 4 position at the cytoplasmic entrance of the channel pore in GluA2, is crucial in polyamine block. Replacing this with a neutral residue reduces the inward rectification caused by channel block (Dingledine et al., 1992; Panchenko et al., 1999). Additionally, neutralization of this Q/R + 4 residue in GluA1 receptors markedly decreases AMPAR single-channel conductance and prevents TARP enhancement of the conductance (Soto et al., 2014). We found that neutralizing the negative charge at position 611 similarly caused a decrease in GluA2(Q) channel conductance. Crucially, GluA2(Q)-D611N conductance was not further attenuated on inclusion of intracellular spermine. It seems likely that D611 plays a key role in ion permeation and its neutralization decreases cation flux. Our results suggest that the presence of positively charged polyamine within the channel may have a similar effect by reducing the impact the negatively charged D611.

The spermine attenuation of CP-AMPAR single-channel conductance was increased by GSG1L and reduced in the presence of TARP γ2. Interestingly, this mirrors the ability of GSG1L and γ2 to, respectively, enhance and relieve polyamine-dependent rectification (Soto et al., 2007; McGee et al., 2015). Indeed, our analyses suggest that GSG1L-mediated reduction in single-channel conductance requires the presence of intracellular polyamines, while the γ2-induced increase in single-channel conductance occurs independently of their presence.

We showed previously that the Q/R + 4 site in GluA1 (D586) plays a crucial role in the enhancement of CP-AMPAR channel conductance by γ2 (Soto et al., 2007; Soto et al., 2014). We now show that this site also determines the effect of GSG1L, as GSG1L does not decrease conductance of GluA2(Q)-D611N. In contrast, GSG1L's attenuation of conductance persisted when the neighboring cysteine residue (Q/R + 3) was replaced (C610S). Both of these residues form part of the AMPAR selectivity filter and likely represent important points of contact for ions in the conduction pathway (Twomey et al., 2019). Although the precise mechanism is unclear, our results suggest that, by reshaping the electronegative cavity between the Q/R and +4 sites, GSG1L hinders polyamine permeation, allowing interactions that decrease the effectiveness of the charged D611 residue as a facilitator of ion transport.

For GluA2(Q) receptors, full-length GSG1L negatively shifted the voltage for half-maximal block by spermine (V_b) and slowed AMPAR recovery from desensitization. GSG1L that lacked the entire C-tail ($GSG1L_{1-228}$) did not affect V_b or recovery. However, a truncated GSG1L that retained 10 membrane-proximal residues of the cytoplasmic C-tail ($GSG1L_{1-238}$) behaved as full-length GSG1L. This suggests that the membrane-proximal region of the GSG1L C-tail is important for its actions. Of note, $GSG1L_{1-228}$ still caused a marked decrease in outward current and a change in RI, suggesting non-C-tail regions of GSG1L also influence polyamine block. Moreover, $GSG1L_{1-228}$ still decreased single-channel conductance and the kinetics of entry into desensitization, indicating that GSG1L's C-tail is not essential in regulating these properties.

The cytoplasmic tails of γ2 and the kainate receptor auxiliary subunits Neto1 and Neto2 have also been shown to modulate polyamine block of AMPARs and kainate receptors, respectively (Fisher and Mott, 2012; Soto et al., 2014). However, unlike GSG1L, these auxiliary subunits reduce rather than enhance spermine block. For Neto1 and Neto2, this process required three positive charges in the C-tail region (Arg, Lys, Lys; Fisher and Mott, 2012). However, in the γ2 C-tail, homologous Arg, His, and Lys residues were not required for the attenuation of polyamine block in CP-AMPARs (Soto et al., 2014), highlighting differences in how auxiliary subunits affect polyamine block. The fact that the proximal C-tail of GSG1L lacks a stretch of equivalent charges suggests that it too acts through a distinct mechanism.

Table 4. Effect of GSG1L and spermine on AMPAR desensitization kinetics

				Compared with GluA alone			
	$[\text{Spermine}]_i$ (mM)	GSG1L (+/−)	Measure	upMD [95% CI]	t	df	p
Homomeric receptors			$\boldsymbol{T_{w,des}}$ **(ms)**				
GluA2(Q)	0.1	−	4.9 ± 0.2 (24)				
	0.1	+	7.8 ± 0.5 (21)	3.0 [2.0, 4.0]	5.90	25.70	<0.0001
	0	−	5.4 ± 0.4 (6)				
	0	+	8.4 ± 0.8 (12)	3.1 [1.6, 4.7]	3.57	15.03	0.0028
GluA4	0.1	−	2.9 ± 0.1 (9)				
	0.1	+	4.3 ± 0.2 (9)	1.4 [1.0, 1.8]	6.25	13.43	<0.0001
	0	−	3.5 ± 0.2 (5)				
	0	+	4.2 ± 0.2 (7)	0.7 [0.2, 1.3]	2.24	8.69	0.053
			$\boldsymbol{T_{w,rec}}$ **(ms)**				
GluA2(Q)	0.1	−	20.7 ± 3.4 (11)				
	0.1	+	322.5 ± 46.5 (21)	302 [225, 403]	6.48	20.22	<0.0001
	0	−	19.3 ± 2.8 (9)				
	0	+	87.1 ± 14.1 (28)	67.8 [45.1, 104]	4.71	28.96	<0.0001
GluA1	0.1	−	183.5 ± 20.1 (14)				
	0.1	+	1,099.1 ± 248.2 (9)	916 [456, 1,380]	3.68	8.10	0.0061
	0	−	145.9 ± 19.5 (7)				
	0	+	278.2 ± 76.8 (8)	132 [−1.5, 286]	1.67	7.89	0.13
GluA4	0.1	−	17.7 ± 2.7 (5)				
	0.1	+	161.2 ± 43.1 (7)	144 [78, 229]	3.32	6.05	0.016
	0	−	28.6 ± 11.7 (3)				
	0	+	24.5 ± 3.7 (6)	−4.0 [−28.1, 12.4]	−0.33	2.42	0.77
Heteromeric receptors			$\boldsymbol{T_{w,des}}$ **(ms)**				
GluA1/A2(R)	0.1	−	5.6 ± 0.3 (15)				
		+	8.0 ± 0.4 (9)	2.4 [1.3, 3.3]	4.47	16.17	0.00038
GluA4/A2(R)	0.1	−	7.0 ± 0.5 (5)				
		+	8.6 ± 0.3 (4)	1.6 [0.4, 2.5]	2.77	6.23	0.031
			$\boldsymbol{T_{w,rec}}$ **(ms)**				
GluA1/A2(R)	0.1	−	72.1 ± 10.7 (11)				
		+	76.3 ± 12.8 (6)	4.2 [−23.3, 37.7]	0.25	11.56	0.80
GluA4/A2(R)	0.1	−	26.5 ± 0.3 (4)				
		+	22.1 ± 1.8 (4)	−4.4 [−12.0, 0.8]	−1.20	4.74	0.29

Summary data presented as mean ± SEM, from (n) patches. The effects of GSG1L with and without added spermine are shown as unpaired mean differences (upMD) compared with GluA alone, with 95% confidence intervals [lower bound, upper bound]. Pairwise tests (against GluA alone) are unpaired two-sided Welch t tests reporting the test statistic (t), degrees of freedom (df), and p values. For entry into desensitization, two-way ANOVA showed no effect of spermine on the action of GSG1L for both GluA2(Q) and GluA4. With GluA2(Q) there was a main effect of GSG1L ($F_{(1,59)} = 44.63$; $p < 0.0001$), no main effect of spermine ($F_{(1,59)} = 1.58$; $p = 0.21$), and no interaction ($F_{(1,59)} = 0.0042$; $p = 0.95$). With GluA4, there was a main effect of GSG1L ($F_{(1,26)} = 36.04$; $p < 0.0001$), no main effect of spermine ($F_{(1,26)} = 1.59$; $p = 0.22$), and no interaction ($F_{(1,26)} = 2.84$; $p = 0.10$). For recovery from desensitization, two-way ANOVA showed that the effect of GSG1L on GluA2(Q), GluA1, and GluA4 was affected by spermine. With GluA2(Q), there was a main effect of GSG1L ($F_{(1,65)} = 39.33$; $p < 0.0001$), no main effect of spermine ($F_{(1,65)} = 0.53$; $p = 0.47$), and an interaction ($F_{(1,65)} = 21.93$; $p < 0.0001$). With GluA1, there was a main effect of GSG1L ($F_{(1,34)} = 6.49$; $p = 0.016$), no main effect of spermine ($F_{(1,34)} = 2.50$; $p = 0.12$), and an interaction ($F_{(1,34)} = 7.97$; $p = 0.0079$). With GluA4, there was no main effect of GSG1L ($F_{(1,17)} = 0.44$; $p = 0.51$), no main effect of spermine ($F_{(1,17)} = 0.025$; $p = 0.88$) but an interaction ($F_{(1,17)} = 9.06$; $p = 0.0079$). For heteromeric receptors, assembly was confirmed by linear current–voltage relationships, giving $RI_{+80/-80}$ values of ∼1.

Early studies showed that GSG1L dramatically slowed AMPAR recovery from desensitization (Schwenk et al., 2012; Shanks et al., 2012). Subsequent work showed that this effect of GSG1L is largely eliminated when its long extracellular β1–β2 loop between TM1 and TM2 (Ex-1 loop) is replaced with the shorter Ex1 loop of γ2 (Twomey et al., 2017). Indeed, a recent study identified an evolutionarily conserved allosteric site within the GluA LBDs (designated the "EK" site) that is assumed to interact with the Ex1 loop of GSG1L and is critical for GSG1L's effect on the recovery from desensitization (Perozzo et al., 2023). Interestingly, replacing the Ex1 domain of γ2 with that of GSG1L fails to confer on γ2 an ability to slow recovery from desensitization, suggesting other elements within the GSG1L scaffold are also needed for this effect (Twomey et al., 2017). Our results suggest the role of the intracellular C-tail of GSG1L in modulating AMPAR recovery from desensitization. Although intriguing, the basis of this effect remains unclear, given that recovery is generally thought to depend largely on rearrangements of extracellular structures, notably AMPAR subunit LBDs (Armstrong et al., 2006; Chen et al., 2017).

Surprisingly, we find that ability of GSG1L to slow recovery from desensitization is greatly diminished when polyamines are absent. In contrast, the slowing of entry into desensitization by GSG1L is polyamine independent. This differential polyamine dependence parallels our finding that the proximal C-tail of GSG1L is required for slowing of recovery from, but not entry into, desensitization. These results could suggest that spermine interacts directly with the proximal region of the GSG1L C-tail to modulate the rate of recovery from desensitization, an effect distinct from its pore blocking action. In this regard, it is interesting to note that GSG1L has been shown to slow recovery from desensitization of heteromeric CI-AMPARs (Schwenk et al., 2012; Perozzo et al., 2023), yet these receptors do not exhibit voltage-dependent block by intracellular spermine (Bowie and Mayer, 1995; Soto et al., 2007).

While biochemical studies suggest that in native receptors GSG1L may preferentially associate with GluA2 (Schwenk et al., 2012; Perozzo et al., 2023), there is indirect evidence for an association with GluA2-lacking CP-AMPARs. Thus, in cerebellar stellate cells, which normally express synaptic CP-AMPARs, transfection with GSG1L has been shown to increase mEPSC rectification (McGee et al., 2015). Conversely, knock down of GSG1L has been shown to enhance EPSC amplitude in hippocampal neurons with artificially elevated synaptic CP-AMPAR expression (McGee et al., 2015) and at a subset of corticothalamic synapses (Kamalova et al., 2020) in nuclei where there is a low expression of GluA2 (Spreafico et al., 1994; Mineff and Weinberg, 2000; Phillips et al., 2019). Although these actions could reflect indirect effects on mixed AMPAR populations, they are parsimoniously explained by direct CP-AMPAR interaction. This issue and the wider functional context in which GSG1L negatively regulates AMPARs remain uncertain.

The free intracellular spermine concentration in neurons has been estimated to be roughly 50 μM (Bowie and Mayer, 1995). In our study, 100 μM added spermine, which likely equates to ∼40 μM free spermine (Soto et al., 2007), attenuated conductance, and slowed the recovery kinetics of CP-AMPARs containing GSG1L. Thus, the concentration of spermine in neurons would be sufficient to modulate the function of synaptic CP-AMPARs containing GSG1L, and this effect would be present at physiologically relevant negative membrane potentials. The effects of spermine could become of particular significance in cases of dysregulated polyamine metabolism (Cason et al., 2003; Casero et al., 2018) deficits in GluA2 Q/R editing (Maas et al., 2001; Peng et al., 2006; Hideyama et al., 2012; Venkatesh et al., 2019) synapse-specific differences in GSG1L prevalence (Kamalova et al., 2020). Although the precise physiological impacts of polyamine/GSG1L interplay remain to be determined, our results suggest that spermine can have effects on AMPAR function beyond open- and closed-channel block.

References

Araneda RC, Lan JY, Zheng X, Zukin RS, Bennett MV (1999) Spermine and arcaine block and permeate *N*-methyl-D-aspartate receptor channels. Biophys J 76:2899–2911.

Armstrong N, Jasti J, Beich-Frandsen M, Gouaux E (2006) Measurement of conformational changes accompanying desensitization in an ionotropic glutamate receptor. Cell 127:85–97.

Bahring R, Bowie D, Benveniste M, Mayer ML (1997) Permeation and block of rat GluR6 glutamate receptor channels by internal and external polyamines. J Physiol 502:575–589.

Bowie D, Mayer ML (1995) Inward rectification of both AMPA and kainate subtype glutamate receptors generated by polyamine-mediated ion channel block. Neuron 15:453–462.

Brown P, McGuire H, Bowie D (2018) Stargazin and cornichon-3 relieve polyamine block of AMPA receptors by enhancing blocker permeation. J Gen Physiol 150:67–82.

Burnashev N, Monyer H, Seeburg PH, Sakmann B (1992) Divalent ion permeability of AMPA receptor channels is dominated by the edited form of a single subunit. Neuron 8:189–198.

Casero Jr RA, Murray ST, Pegg AE (2018) Polyamine metabolism and cancer: treatments, challenges and opportunities. Nat Rev Cancer 18:681–695.

Cason AL, et al. (2003) X-linked spermine synthase gene (SMS) defect: the first polyamine deficiency syndrome. Eur J Hum Genet 11:937–944.

Certain N, Gan Q, Bennett J, Hsieh H, Wollmuth LP (2023) Differential regulation of tetramerization of the AMPA receptor glutamate-gated ion channel by auxiliary subunits. J Biol Chem 299:105227.

Chen S, Zhao Y, Wang Y, Shekhar M, Tajkhorshid E, Gouaux E (2017) Activation and desensitization mechanism of AMPA receptor-TARP complex by cryo-EM. Cell 170:1234–1246. e1214.

Cho CH, St-Gelais F, Zhang W, Tomita S, Howe JR (2007) Two families of TARP isoforms that have distinct effects on the kinetic properties of AMPA receptors and synaptic currents. Neuron 55:890–904.

Coombs ID, Cull-Candy SG (2021) Single-channel mechanisms underlying the function, diversity and plasticity of AMPA receptors. Neuropharmacology 198:108781.

Coombs ID, Soto D, Zonouzi M, Renzi M, Shelley C, Farrant M, Cull-Candy SG (2012) Cornichons modify channel properties of recombinant and glial AMPA receptors. J Neurosci 32:9796–9804.

Cull-Candy SG, Farrant M (2021) Ca^{2+}-permeable AMPA receptors and their auxiliary subunits in synaptic plasticity and disease. J Physiol 599:2655–2671.

Dingledine R, Hume RI, Heinemann SF (1992) Structural determinants of barium permeation and rectification in non-NMDA glutamate receptor channels. J Neurosci 12:4080–4087.

Donevan SD, Rogawski MA (1995) Intracellular polyamines mediate inward rectification of Ca^{2+}-permeable α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptors. Proc Natl Acad Sci U S A 92:9298–9302.

Fisher JL, Mott DD (2012) The auxiliary subunits Neto1 and Neto2 reduce voltage-dependent inhibition of recombinant kainate receptors. J Neurosci 32:12928–12933.

Fleidervish IA, Libman L, Katz E, Gutnick MJ (2008) Endogenous polyamines regulate cortical neuronal excitability by blocking voltage-gated Na^+ channels. Proc Natl Acad Sci U S A 105:18994–18999.

Green WN, Andersen OS (1991) Surface charges and ion channel function. Annu Rev Physiol 53:341–359.

Greger IH, Watson JF, Cull-Candy SG (2017) Structural and functional architecture of AMPA-type glutamate receptors and their auxiliary proteins. Neuron 94:713–730.

Gu X, Mao X, Lussier MP, Hutchison MA, Zhou L, Hamra FK, Roche KW, Lu W (2016) GSG1L suppresses AMPA receptor-mediated synaptic transmission and uniquely modulates AMPA receptor kinetics in hippocampal neurons. Nat Commun 7:10873.

Guo D, Lu Z (2000) Mechanism of IRK1 channel block by intracellular polyamines. J Gen Physiol 115:799–814.

Haghighi AP, Cooper E (1998) Neuronal nicotinic acetylcholine receptors are blocked by intracellular spermine in a voltage-dependent manner. J Neurosci 18:4050–4062.

Haghighi AP, Cooper E (2000) A molecular link between inward rectification and calcium permeability of neuronal nicotinic acetylcholine α3β4 and α4β2 receptors. J Neurosci 20:529–541.

Hansen KB, et al. (2021) Structure, function, and pharmacology of glutamate receptor ion channels. Pharmacol Rev 73:298–487.

Henley JM, Wilkinson KA (2016) Synaptic AMPA receptor composition in development, plasticity and disease. Nat Rev Neurosci 17:337–350.

Hideyama T, Yamashita T, Aizawa H, Tsuji S, Kakita A, Takahashi H, Kwak S (2012) Profound downregulation of the RNA editing enzyme ADAR2 in ALS spinal motor neurons. Neurobiol Dis 45:1121–1128.

Jackson AC, Nicoll RA (2011) Stargazin (TARP γ-2) is required for compartment-specific AMPA receptor trafficking and synaptic plasticity in cerebellar stellate cells. J Neurosci 31:3939–3952.

Jogini V, Jensen MO, Shaw DE (2023) Gating and modulation of an inward-rectifier potassium channel. J Gen Physiol 155:e202213085.

Kamalova A, Futai K, Delpire E, Nakagawa T (2020) AMPA receptor auxiliary subunit GSG1L suppresses short-term facilitation in corticothalamic synapses and determines seizure susceptibility. Cell Rep 32:107921.

Kamboj SK, Swanson GT, Cull-Candy SG (1995) Intracellular spermine confers rectification on rat calcium-permeable AMPA and kainate receptors. J Physiol 486:297–303.

Kim J, Moon SH, Shin YC, Jeon JH, Park KJ, Lee KP, So I (2016) Intracellular spermine blocks TRPC4 channel via electrostatic interaction with C-terminal negative amino acids. Pflugers Arch 468:551–561.

Koh DS, Burnashev N, Jonas P (1995) Block of native Ca(2+)-permeable AMPA receptors in rat brain by intracellular polyamines generates double rectification. J Physiol 486:305–312.

Kuner T, Beck C, Sakmann B, Seeburg PH (2001) Channel-lining residues of the AMPA receptor M2 segment: structural environment of the Q/R site and identification of the selectivity filter. J Neurosci 21:4162–4172.

Lopatin AN, Makhina EN, Nichols CG (1994) Potassium channel block by cytoplasmic polyamines as the mechanism of intrinsic rectification. Nature 372:366–369.

Lu Z, Ding L (1999) Blockade of a retinal cGMP-gated channel by polyamines. J Gen Physiol 113:35–43.

Maas S, Patt S, Schrey M, Rich A (2001) Underediting of glutamate receptor GluR-B mRNA in malignant gliomas. Proc Natl Acad Sci U S A 98:14687–14692.

Maksaev G, Yuan P, Nichols CG (2023) Blockade of TRPV channels by intracellular spermine. J Gen Physiol 155:e202213273.

McGee TP, Bats C, Farrant M, Cull-Candy SG (2015) Auxiliary subunit GSG1L acts to suppress calcium-permeable AMPA receptor function. J Neurosci 35:16171–16179.

Mineff EM, Weinberg RJ (2000) Differential synaptic distribution of AMPA receptor subunits in the ventral posterior and reticular thalamic nuclei of the rat. Neuroscience 101:969–982.

Neher E, Steinbach JH (1978) Local anaesthetics transiently block currents through single acetylcholine-receptor channels. J Physiol 277:153–176.

Oliver D, Baukrowitz T, Fakler B (2000) Polyamines as gating molecules of inward-rectifier K^+ channels. Eur J Biochem 267:5824–5829.

Panchenko VA, Glasser CR, Mayer ML (2001) Structural similarities between glutamate receptor channels and K^+ channels examined by scanning mutagenesis. J Gen Physiol 117:345–360.

Panchenko VA, Glasser CR, Partin KM, Mayer ML (1999) Amino acid substitutions in the pore of rat glutamate receptors at sites influencing block by polyamines. J Physiol 520:337–357.

Peng PL, Zhong X, Tu W, Soundarapandian MM, Molner P, Zhu D, Lau L, Liu S, Liu F, Lu Y (2006) ADAR2-dependent RNA editing of AMPA receptor subunit GluR2 determines vulnerability of neurons in forebrain ischemia. Neuron 49:719–733.

Perozzo AM, Schwenk J, Kamalova A, Nakagawa T, Fakler B, Bowie D (2023) GSG1L-containing AMPA receptor complexes are defined by their spatiotemporal expression, native interactome and allosteric sites. Nat Commun 14:6799.

Phillips JW, et al. (2019) A repeated molecular architecture across thalamic pathways. Nat Neurosci 22:1925–1935.

Price MG, Davis CF, Deng F, Burgess DL (2005) The α-amino-3-hydroxyl-5-methyl-4-isoxazolepropionate receptor trafficking regulator “stargazin” is related to the claudin family of proteins by Its ability to mediate cell-cell adhesion. J Biol Chem 280:19711–19720.

Pusch M, Bertorello L, Conti F (2000) Gating and flickery block differentially affected by rubidium in homomeric KCNQ1 and heteromeric KCNQ1/KCNE1 potassium channels. Biophys J 781:211–226.

Ramos-Vicente D, Grant SG, Bayes A (2021) Metazoan evolution and diversity of glutamate receptors and their auxiliary subunits. Neuropharmacology 195:108640.

Rock DM, MacDonald RL (1992) Spermine and related polyamines produce a voltage-dependent reduction of *N*-methyl-D-aspartate receptor single-channel conductance. Mol Pharmacol 42:157–164.

Rock DM, Macdonald RL (1995) Polyamine regulation of N-methyl-D-aspartate receptor channels. Annu Rev Pharmacol Toxicol 35:463–482.

Rothman JS, Silver RA (2018) Neuromatic: an integrated open-source software toolkit for acquisition, analysis and simulation of electrophysiological data. Front Neuroinform 12:14.

Rouach N, Byrd K, Petralia RS, Elias GM, Adesnik H, Tomita S, Karimzadegan S, Kealey C, Bredt DS, Nicoll RA (2005) TARP γ-8 controls hippocampal AMPA receptor number, distribution and synaptic plasticity. Nat Neurosci 8:1525–1533.

Schwenk J, et al. (2012) High-resolution proteomics unravel architecture and molecular diversity of native AMPA receptor complexes. Neuron 74:621–633.

Schwenk J, et al. (2019) An ER assembly line of AMPA-receptors controls excitatory neurotransmission and its plasticity. Neuron 104:680–692. e689.

Shanks NF, et al. (2012) Differences in AMPA and kainate receptor interactomes facilitate identification of AMPA receptor auxiliary subunit GSG1L. Cell Rep 1:590–598.

Sommer B, Kohler M, Sprengel R, Seeburg PH (1991) RNA editing in brain controls a determinant of ion flow in glutamate-gated channels. Cell 67:11–19.

Soto D, Coombs ID, Kelly L, Farrant M, Cull-Candy SG (2007) Stargazin attenuates intracellular polyamine block of calcium-permeable AMPA receptors. Nat Neurosci 10:1260–1267.

Soto D, Coombs ID, Gratacos-Batlle E, Farrant M, Cull-Candy SG (2014) Molecular mechanisms contributing to TARP regulation of channel conductance and polyamine block of calcium-permeable AMPA receptors. J Neurosci 34:11673–11683.

Spreafico R, Frassoni C, Arcelli P, Battaglia G, Wenthold RJ, De Biasi S (1994) Distribution of AMPA selective glutamate receptors in the thalamus of adult rats and during postnatal development. a light and ultrastructural immunocytochemical study. Brain Res Dev Brain Res 82:231–244.

Swanson GT, Kamboj SK, Cull-Candy SG (1997) Single-channel properties of recombinant AMPA receptors depend on RNA editing, splice variation, and subunit composition. J Neurosci 17:58–69.

Tomita S, Adesnik H, Sekiguchi M, Zhang W, Wada K, Howe JR, Nicoll RA, Bredt DS (2005) Stargazin modulates AMPA receptor gating and trafficking by distinct domains. Nature 435:1052–1058.

Twomey EC, Yelshanskaya MV, Grassucci RA, Frank J, Sobolevsky AI (2017) Channel opening and gating mechanism in AMPA-subtype glutamate receptors. Nature 549:60–65.

Twomey EC, Yelshanskaya MV, Vassilevski AA, Sobolevsky AI (2018) Mechanisms of channel block in calcium-permeable AMPA receptors. Neuron 99:956–968. e954.

Twomey EC, Yelshanskaya MV, Sobolevsky AI (2019) Structural and functional insights into transmembrane AMPA receptor regulatory protein complexes. J Gen Physiol 151:1347–1356.

Venkatesh HS, et al. (2019) Electrical and synaptic integration of glioma into neural circuits. Nature 573:539–545.

Verdoorn TA, Burnashev N, Monyer H, Seeburg PH, Sakmann B (1991) Structural determinants of ion flow through recombinant glutamate receptor channels. Science 252:1715–1718.

Watanabe S, Kusama-Eguchi K, Kobayashi H, Igarashi K (1991) Estimation of polyamine binding to macromolecules and ATP in bovine lymphocytes and rat liver. J Biol Chem 266:20803–20809.

Wilding TJ, Chen K, Huettner JE (2010) Fatty acid modulation and polyamine block of GluK2 kainate receptors analyzed by scanning mutagenesis. J Gen Physiol 136:339–352.

Xie LH, John SA, Weiss JN (2002) Spermine block of the strong inward rectifier potassium channel Kir2.1: dual roles of surface charge screening and pore block. J Gen Physiol 120:53–66.

Yang Y, Sigworth FJ (1998) Single-channel properties of IKs potassium channels. J Gen Physiol 112:665–678.

Yellen G (1984) Relief of Na^+ block of Ca^{2+}-activated K^+ channels by external cations. J Gen Physiol 84:187–199.

Zhang Y, Niu X, Brelidze TI, Magleby KL (2006) Ring of negative charge in BK channels facilitates block by intracellular Mg^{2+} and polyamines through electrostatics. J Gen Physiol 128:185–202.